Lecture 23: Characteristic Functions and Limit Theorems

Probability II Notes

1 Characteristic Functions

Definition 1. *Let X be a random variable (r.v.). The characteristic function of X, denoted by $\phi_X(t)$, is defined as:*

$$\phi_X(t) = \mathbb{E}[e^{itX}], \quad t \in \mathbb{R}, \quad i^2 = -1 \tag{1}$$

Expanding this using Euler's formula:

$$\phi_X(t) = \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)]$$

2 Example: Normal Distribution

Let $X \sim N(0, 1)$. The PDF is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad x \in \mathbb{R}$$

Calculating the characteristic function:

$$\begin{aligned} \phi_X(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{itx} e^{-x^2/2} dx \\ &= \frac{e^{-t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-it)^2} dx \end{aligned}$$

The integral component resolves to $\sqrt{2\pi}$ (using standard Gaussian integration techniques/contour integration). Thus:

$$\phi_X(t) = e^{-t^2/2} \tag{2}$$

3 A Different Path: Series Expansion

We can derive the same result using the Taylor series expansion of the exponential function.

$$e^{itX} = 1 + \frac{itX}{1!} + \frac{(itX)^2}{2!} + \frac{(itX)^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{(itX)^n}{n!}$$

Taking the expectation:

$$\phi_X(t) = \mathbb{E}\left[\sum_{n=0}^{\infty} \frac{(itX)^n}{n!}\right] = \sum_{n=0}^{\infty} \frac{(it)^n}{n!}\mathbb{E}[X^n]$$

We use the known moments of the Standard Normal distribution $X \sim N(0,1)$:

$$\mathbb{E}[X^n] = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \frac{(2m)!}{2^m m!} & \text{if } n = 2m \text{ (even)} \end{cases}$$

Substituting this back into the series, the odd terms vanish. We sum only over $n = 2m$:

$$\begin{aligned}\phi_X(t) &= \sum_{m=0}^{\infty} \frac{(it)^{2m}}{(2m)!}\mathbb{E}[X^{2m}] \\ &= \sum_{m=0}^{\infty} \frac{(it)^{2m}}{(2m)!}\frac{(2m)!}{2^m m!} \\ &= \sum_{m=0}^{\infty} i^{2m}\frac{t^{2m}}{2^m m!}\end{aligned}$$

Since $i^{2m} = (i^2)^m = (-1)^m$, we get:

$$\phi(t) = \sum_{m=0}^{\infty}(-1)^m \left(\frac{t^2}{2}\right)^m \frac{1}{m!}$$

Recognizing the Maclaurin series for e^x, this sums to:

$$\phi_X(t) = e^{-t^2/2}, \quad t \in \mathbb{R}$$

4 Central Limit Theorem (Lindeberg-Lévy)

Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ and variance $\sigma^2 > 0$. Let $S_n = \sum_{i=1}^n X_i$. Then:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1) \quad \text{as } n \to \infty$$

4.1 Lévy's Continuity Theorem

This theorem allows us to relate convergence of characteristic functions to convergence in distribution.

Theorem 1. *Let $(Z_n)_{n\geq 1}$ be a sequence of random variables with characteristic functions $(\phi_n)_n$, and let Z be a random variable with characteristic function ϕ.*

1. *If $Z_n \xrightarrow{d} Z$, then $\phi_n(t) \to \phi(t)$ for all $t \in \mathbb{R}$.*
2. *Conversely, if $\phi_n(t) \to \psi(t)$ for all $t \in \mathbb{R}$, where $\psi(t)$ is continuous at $t = 0$, then there exists a random variable Z such that $\phi_Z = \psi$ and $Z_n \xrightarrow{d} Z$.*

4.2 Uniqueness Theorem

If F and G are two distribution functions:

$$\phi_F(t) = \phi_G(t) \implies F = G$$

There is a homomorphism between the set of all distributions and the set of all characteristic functions.

5 Proof of the CLT

We wish to show:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1)$$

Step 1: Standardization Without loss of generality, we assume $\mu = 0$ and $\sigma^2 = 1$. If not, we use the transformation $X_i \to \frac{X_i - \mu}{\sigma}$. Thus, assume:

$$\mathbb{E}[X_i] = 0 \quad \text{and} \quad \mathbb{E}[X_i^2] = 1, \quad \forall i \geq 1$$

We want to show that $\frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0,1)$. By Lévy's Continuity Theorem, it suffices to show:

$$\phi_{S_n/\sqrt{n}}(t) \to e^{-t^2/2}, \quad \forall t \in \mathbb{R}$$

Step 2: Characteristic Function Setup

$$\begin{aligned}
\phi_{S_n/\sqrt{n}}(t) &= \mathbb{E}\left[e^{it\frac{S_n}{\sqrt{n}}}\right] \\
&= \mathbb{E}\left[e^{i\frac{t}{\sqrt{n}}(X_1+\cdots+X_n)}\right] \\
&= \mathbb{E}\left[\prod_{j=1}^{n} e^{i\frac{t}{\sqrt{n}}X_j}\right]
\end{aligned}$$

Since $(X_k)_{k\geq 1}$ are i.i.d., this becomes:

$$= \left(\mathbb{E}\left[e^{i\frac{t}{\sqrt{n}}X_1}\right]\right)^n$$

Step 3: Taylor Expansion We expand the inner expectation using the series definition:

$$\begin{aligned}
\mathbb{E}\left[e^{i\frac{t}{\sqrt{n}}X_1}\right] &= \mathbb{E}\left[\sum_{m=0}^{\infty} \frac{(it/\sqrt{n})^m X_1^m}{m!}\right] \\
&= 1 + \frac{it}{\sqrt{n}}\mathbb{E}[X_1] + \frac{(it)^2}{2n}\mathbb{E}[X_1^2] + o\left(\frac{1}{n}\right)
\end{aligned}$$

Substituting $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = 1$:

$$= 1 + 0 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)$$

Step 4: The Limit We now have:

$$\phi_{S_n/\sqrt{n}}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n$$

Using the fundamental limit property:

$$\left(1 + \frac{\lambda}{n} + a_n\right)^n \to e^{\lambda} \quad \text{as } n \to \infty \text{ if } na_n \to 0$$

Here, $\lambda = -t^2/2$. Thus:

$$\lim_{n\to\infty} \phi_{S_n/\sqrt{n}}(t) = e^{-t^2/2}$$

Detailed Error Analysis (Remainder Term)

To be rigorous about the remainder term $a_n(t)$, we consider the expansion further:

$$\mathbb{E}\left[e^{i\frac{t}{\sqrt{n}}X}\right] = 1 + \frac{it}{\sqrt{n}}\mathbb{E}[X] - \frac{t^2}{2n}\mathbb{E}[X^2] + a_n(t)$$

Where the remainder $a_n(t)$ sums the higher order terms ($m \geq 3$):

$$a_n(t) = \sum_{m=3}^{\infty} \frac{(it/\sqrt{n})^m}{m!}\mathbb{E}[X^m]$$

Factoring out $(it/\sqrt{n})^3$:

$$a_n(t) = \frac{(it)^3}{n^{3/2}} \sum_{k=0}^{\infty} \frac{(it/\sqrt{n})^k}{(k+3)!}\mathbb{E}[X^{k+3}]$$

Let $g_n(t)$ represent the summation part. Then:

$$n \cdot a_n(t) = n \cdot \frac{t^3}{n^{3/2}} g_n(t) = \frac{t^3}{\sqrt{n}} g_n(t)$$

As $n \to \infty$, $\frac{1}{\sqrt{n}} \to 0$. Assuming bounded moments, the term vanishes. Thus, the limit holds:

$$\phi_n(t) \to e^{-t^2/2}$$

This confirms convergence to the Standard Normal Distribution $N(0, 1)$.